🎙️ KisanMitra AI — Balanced 50/50 Live Demo Script

A Perfectly Divided 2-Member Presentation for Evaluators & Teachers

> ***Structure:*** *Exactly 50% speaking time and demo actions for each member. -* ***Member 1 (Akash):*** *Intro, Problem, Hinglish Text NLU, Mandi Live Data, Follow-up Memory, and NLP Inspector Sheet. -* ***Member 2 (Partner):*** *Handover, Hyperlocal Weather & Farming Tips, Hands-Free Voice Mode, AI Leaf Doctor, and Conclusion.*

🟢 PART 1: Presented by Member 1 (Akash)

(Duration: ~4 Minutes)

1. Introduction & The Problem

👤 **Member 1 (Akash):**

> *"Good morning respected teachers and friends! My name is Akash, and together with my team member, we have built KisanMitra AI—an intelligent, multilingual agricultural assistant designed specifically for Indian farmers.*
>
> *Today, millions of Indian farmers have smartphones, but agricultural apps fail them because they are either written in formal English or require navigating complex menus. In reality, a farmer speaks Marathi or Hindi, or types in Latin letters—what we call Hinglish.*
>
> *Our goal was to create a conversational assistant that understands text, voice, and leaf photos—running entirely on free APIs and local on-device AI."*

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2. Live Demo: Hinglish & Language Intelligence

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🖥️ **Live Action:** Member 1 opens the chat and types:

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`mera tamatar ka paudha peela ho raha hai, kya karu?`

👤 **Member 1 (Akash):**

> *"Let's test our first core feature: Linguistic Intelligence.*
>
> *Notice I didn't use Devanagari script. I typed in casual Hinglish: 'mera tamatar ka paudha peela ho raha hai'. Let's hit send.*
>
> *[Bot replies immediately in Romanized Hindi]*
>
> *In under a second, KisanMitra identified the crop as Tomato, diagnosed Yellowing leaves, provided treatment advice, and gave the official Kisan Call Centre helpline (1800-180-1551).*
>
> *Most off-the-shelf NLP detectors fail on Hinglish because they mistake it for Italian or English. Our backend uses character n-grams (ranges 2 to 5) with Logistic Regression. So whether a farmer types 'pyaz', 'pyaj', or 'pyaaz', the model captures the subword roots and understands the exact intent!"*

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3. Live Demo: Live Mandi Rates & Conversational Memory

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🖥️ **Live Action:** Member 1 types:

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`Nashik me pyaz ka bhav kya hai?`

👤 **Member 1 (Akash):**

> *"Now let's check market prices. I'm asking: 'What is the onion price in Nashik?'*
>
> *[Bot renders the live Mandi Card]*
>
> *This isn't hardcoded dummy data. KisanMitra connects directly to the Government of India's Agmarknet API on data.gov.in, giving today's actual modal price per quintal.*
>
> *Now, watch what happens in a real conversation when a farmer asks a follow-up. I won't repeat the question. I will just type three words:"*

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🖥️ **Live Action:** Member 1 types only 3 words:

**

`aur Pune me?`

**

🖥️ **Live Action:** Bot gives Onion prices in Pune!

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👤 **Member 1 (Akash):**

> *"Notice I never said 'onion' or 'price'! Our system has Dialogue State Tracking. It remembers that the previous topic was Onion Prices, connects 'Pune' to it, and answers instantly without losing context."*

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4. Behind-the-Scenes: The NLP Inspector Sheet

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🖥️ **Live Action:** Member 1 clicks the 🧠 Brain icon on the message.

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👤 **Member 1 (Akash):**

> *"For our teachers and evaluators, we added a transparency feature: clicking this small brain icon opens our NLP Inspector Sheet.*
>
> *It reveals exactly what the model decided behind the scenes: - Detected Language: Hindi (Romanized) - Predicted Intent:* `market_price` *with 98% confidence - Extracted Slots: Crop = Onion, Location = Pune - Follow-up Memory: Active*
>
> *Now, I will hand over to my partner to show you our weather engine, hands-free voice mode, and on-device AI leaf diagnosis!"*

🔵 PART 2: Presented by Member 2 (Partner)

(Duration: ~4 Minutes)

5. Hyperlocal Weather & Agro-Advisory

👥 **Member 2 (Partner):**

> *"Thank you, Akash! Respected teachers, I will now walk you through our real-time weather system, voice interaction, and computer vision model.*
>
> *Farmers don't just want temperature numbers; they need to know what the weather means for their crops. Let's see how KisanMitra handles this."*

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🖥️ **Live Action:** Member 2 clicks the location pin in the sidebar or 'Use my current location' GPS.

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👥 **Member 2 (Partner):**

> *"

🖥️ **Live Action:** Weather card loads with 5-day forecast

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> *KisanMitra supports 1-tap browser GPS, which reverse-geocodes your coordinates into the exact village and district.*
>
> *Notice the agronomic tip at the bottom of the card: 'Rain is likely in the next 2 days: postpone pesticide spraying and fertilizer application; ensure fields drain well.'*
>
> *We programmed smart agricultural heuristics into the weather engine: if high winds are predicted, it warns against chemical spray drift; if temperatures drop near freezing, it gives frost protection tips."*

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6. Live Demo: Hands-Free Voice Mode

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🖥️ **Live Action:** Member 2 clicks the large Microphone 🎙️ button. The glowing animated Voice Orb appears.

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👥 **Member 2 (Partner):**

> *"Now, what if a farmer cannot read or type? We built a full Hands-Free Voice Mode.*
>
> *Watch as I speak to it in Hindi:"*

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🖥️ **Live Action:** Member 2 speaks clearly into the mic in Hindi:

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> *"इंदौर में गेहूं का भाव क्या है?"*
> *(What is the price of wheat in Indore?)*

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🖥️ **Live Action:** Visualizer pulses, transcribes the voice, and the bot speaks the answer out loud in natural Hindi voice.

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👥 **Member 2 (Partner):**

> *"Let's break down what just happened in under 2 seconds: 1. Our local faster-whisper AI detected the spoken language directly from audio waves. 2. Google's Speech-to-Text transcribed the words using our custom farming vocabulary hints. 3. Our pipeline found the wheat price in Indore. 4. And text-to-speech spoke the answer out loud in natural Hindi! Complete hands-free communication!"*

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7. Live Demo: The AI Crop Leaf Doctor

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🖥️ **Live Action:** Member 2 clicks + next to the chat box → selects or uploads a photo of a diseased leaf (e.g., Tomato early blight).

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👥 **Member 2 (Partner):**

> *"Our final and most powerful feature is The AI Crop Leaf Doctor.*
>
> *When a farmer spots unusual leaf spots or rotting in the field, they can simply snap a photo with their phone camera or upload it."*

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🖥️ **Live Action:** Member 2 clicks Send. In under 1 second, the Diagnosis Card appears showing: 'Tomato — Early Blight (94% Confident)' with treatment steps.

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👥 **Member 2 (Partner):**

> *"Notice how fast that was! Most apps send images to expensive cloud vision APIs. Instead, we fine-tuned a MobileNetV2 neural network on the 38-class PlantVillage dataset and converted it to an ONNX runtime.*
>
> *It is only 9 MB in size and runs directly on the CPU in under 40 milliseconds! It evaluates 4 different angles of the leaf, identifies the exact plant pathology, and provides practical organic and chemical remedies."*

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8. Conclusion & Impact

👥 **Member 2 (Partner):**

> *"To conclude our presentation: - Member 1 (Akash) showed our NLU pipeline handling Hinglish, live government Mandi rates, and multi-turn conversational memory. - I demonstrated our hyperlocal weather engine, hands-free voice mode, and on-device plant disease diagnosis.*
>
> *KisanMitra AI proves that cutting-edge AI and NLP can be brought to rural India using 100% free APIs and efficient on-device models.*
>
> *Thank you very much! Akash and I are now ready to answer any questions from the panel."*

📋 Member Checklist for the Demo:

Part	Presenter	Key Feature Demonstrated	Technical Highlight Mentioned
Part 1	**Member 1 (Akash)**	Problem intro, Hinglish query, Tomato yellowing advice	Char n-gram (2–5) TF-IDF + Logistic Regression
		Mandi rates check, follow-up query (*"aur Pune me?"*)	Live Agmarknet data.gov.in API, Dialogue State Tracking
		Brain 🧠 icon inspection sheet	Explainable NLP (confidence score, slots, intent)
Part 2	**Member 2 (Partner)**	GPS / District weather forecast + farming tips	Dual-provider weather + agronomic heuristics
		Hands-free animated voice mode (speaks in Hindi)	Whisper audio LID + Google STT + gTTS
		Leaf photo upload → Early Blight diagnosis	MobileNetV2 ONNX (~9 MB, 40ms CPU inference)
		Project summary & conclusion	Zero-budget rural AI architecture